



North Sydney Girls High School

2022

HSC TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 10 minutes
- Working Time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
70

Section I – 10 marks (pages 3 – 6)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 7 – 13)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER:

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Question	1-10	11	12	13	14	Total
Mark	/10	/15	/15	/15	/15	/70

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1-10.

- 1 Let $P(x) = x^3 - 2ax^2 + x - 1$ where $a \in \mathbb{R}$. When $P(x)$ is divided by $x + 2$, the remainder is 5. What is the value of a ?
- A. 2
- B. $-\frac{7}{4}$
- C. $\frac{1}{2}$
- D. -2
- 2 The points A and B have coordinates $(-2, 3)$ and $(2, -5)$ respectively. Which of the following is the vector $\overrightarrow{AB}$?
- A. $-2\hat{j}$
- B. $4\hat{i} - 8\hat{j}$
- C. $-4\hat{i} + 8\hat{j}$
- D. $2\hat{j}$
- 3 What is the angle between the vectors $\begin{pmatrix} -7 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$?
- A. $\cos^{-1}(-0.8)$
- B. $\cos^{-1}(-0.08)$
- C. $\cos^{-1}(0.8)$
- D. $\cos^{-1}(0.08)$

4 Which of the following is the derivative of $\tan^{-1}(3x)$?

A. $3\tan^{-1} 3x$

B. $\frac{3}{1+9x^2}$

C. $\frac{3}{1+3x^2}$

D. $3\sec^2 3x$

5 What is the equation of the inverse of $f(x) = \frac{5+e^{2x}}{3}$?

A. $y = \frac{3}{5+e^{2x}}$

B. $y = e^{5-3x}$

C. $y = \frac{1}{2} \ln(3x-5)$

D. $y = \frac{1}{2} \ln(5-3x)$

6 Four female and four male students are to be seated around a circular table. In how many ways can this be done if the males and females must alternate?

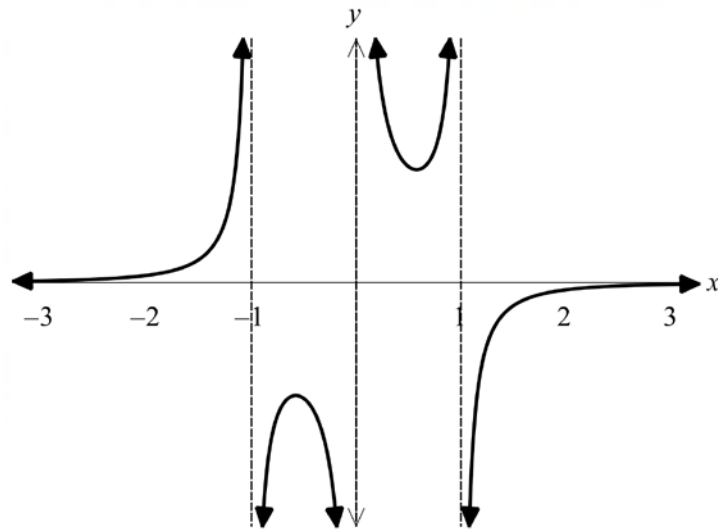
A. $4! \times 4!$

B. $3! \times 4!$

C. $3! \times 3!$

D. $2 \times 3! \times 3!$

- 7 The graph below shows $y = \frac{1}{f(x)}$.



Which of the following best represents the equation of $f(x)$?

- A. $f(x) = 1 - x^2$
- B. $f(x) = x(x^2 - 1)$
- C. $f(x) = x(1 - x^2)$
- D. $f(x) = x^2(x^2 - 1)$
- 8 What is the vector projection of $\underline{a} = 2\underline{i} + 3\underline{j}$ in the direction of $\underline{b} = \underline{i} - 4\underline{j}$?

- A. $-\frac{20}{17}\underline{i} - \frac{30}{17}\underline{j}$
- B. $-\frac{10}{13}\underline{i} + \frac{40}{13}\underline{j}$
- C. $-\frac{20}{13}\underline{i} - \frac{30}{13}\underline{j}$
- D. $-\frac{10}{17}\underline{i} + \frac{40}{17}\underline{j}$

- 9** The radius of a sphere, r , is increasing at the rate of 0.3 cm per second.
What is the rate of increase in the volume, V , in cm^3 per second, at the instant when the surface area is $100\pi \text{ cm}^2$?
- A. 10π
B. 12π
C. 25π
D. 30π
- 10** Which of the following is the range of the function $f(x) = |b \cos^{-1}(x) - a|$, where $a > 0$, $b > 0$ and $a < \frac{b\pi}{2}$?
- A. $[-a, b\pi - a]$
B. $[0, b\pi - a]$
C. $[a, b\pi - a]$
D. $[0, a]$

End of Section I

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) Solve $|2x - 3| \leq 1$. 2

(b) Find $\int_0^{\frac{1}{2}} \frac{dy}{\sqrt{1-3y^2}}$. 2

(c) Let α , β and γ be the roots of the equation $2x^3 - kx^2 - 4x + 12 = 0$.

(i) Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. 2

(ii) Given that two of its roots sum to zero, find the third root and hence find the value of k . 2

(d) Using the substitution $t = \tan \frac{\theta}{2}$, or otherwise, show that 2

$$\cot \theta + \frac{1}{2} \tan \frac{\theta}{2} = \frac{1}{2} \cot \frac{\theta}{2} \text{ for all } \theta \neq k\pi, k \in \mathbb{Z}.$$

(e) Find the term independent of x in the expansion of $\left(3x^2 + \frac{2}{x}\right)^{12}$. 2

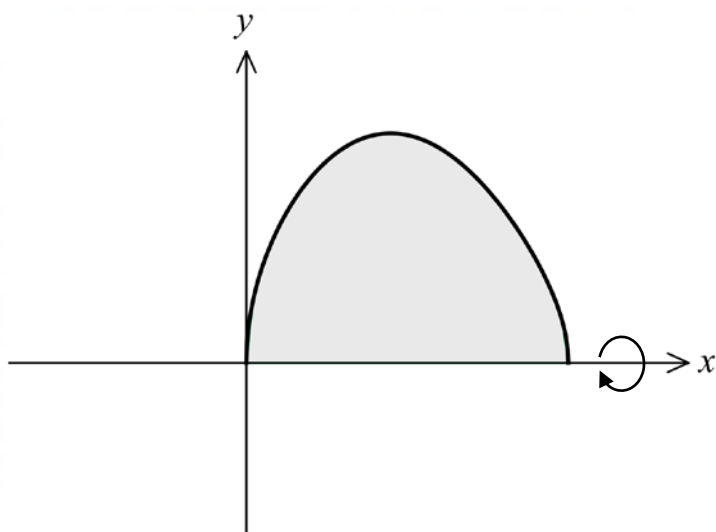
(f) Prove by mathematical induction that $n^3 + 2n$ is divisible by 3 for all positive integers n . 3

Question 12 (15 marks) Use a SEPARATE writing booklet

(a) Solve $\frac{x^2 + 6}{x} < 5$. **3**

(b) By expressing $\cos x - \sqrt{3} \sin x$ in the form $A \cos(x + \alpha)$ where $A > 0$, solve $\cos x - \sqrt{3} \sin x + 1 = 0$ for $0 \leq x \leq 2\pi$. **4**

(c) A section of the graph of $y = \sqrt{\sin 3x \cos 2x}$ is shown in the diagram below. **4**

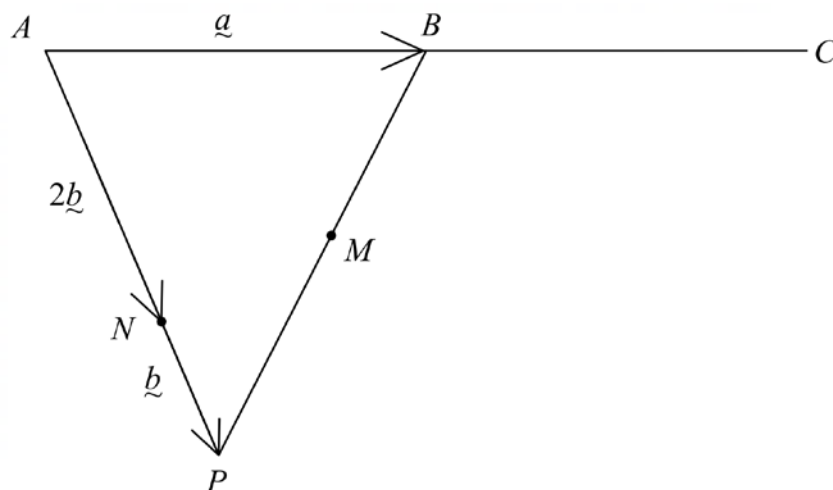


By first finding the smallest positive solution to $\sin 3x \cos 2x = 0$, find the volume of the solid formed when the shaded region is rotated about the x -axis.

Question 12 continues on page 9

Question 12 (continued)

- (d) In the diagram below APB is a triangle. N is a point on AP .
 $\overrightarrow{AB} = \underline{a}$ $\overrightarrow{AN} = 2\underline{b}$ $\overrightarrow{NP} = \underline{b}$



- (i) Find the vector $\overrightarrow{PB}$ in terms of $\underline{a}$ and $\underline{b}$. 1
- (ii) B is the midpoint of AC . M is the midpoint of PB .
 Show that NMC is a straight line. 3

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) A netball team's record for the 2022 season was 16 wins and 4 losses. 2
None of their games were drawn. Prove that the team must have won at least 4 games in a row somewhere during the season.

- (b) The letters of the word REORDER are arranged randomly in a line.

- (i) Use a combinatorial argument to explain why 2

$$\binom{7}{3}\binom{4}{2}\binom{2}{1}\binom{1}{1} = \binom{7}{1}\binom{6}{1}\binom{5}{2}\binom{3}{3}.$$

- (ii) Hence, or otherwise, find the probability that a random rearrangement has all the consonants grouped together. 3

- (c) A pilot is performing at an air show. The position of her aeroplane at time t relative to a fixed origin O is given by $\underline{r}(t) = \left(450 - 150\sin\left(\frac{\pi t}{6}\right)\right)\underline{i} + \left(400 - 200\cos\left(\frac{\pi t}{6}\right)\right)\underline{j}$, where $\underline{i}$ is a unit vector in a horizontal direction and $\underline{j}$ is a unit vector vertically up. Displacement components are measured in metres and time t is measured in seconds where $t \geq 0$.

- (i) Show that the cartesian equation of the path of the aeroplane is given by: 2

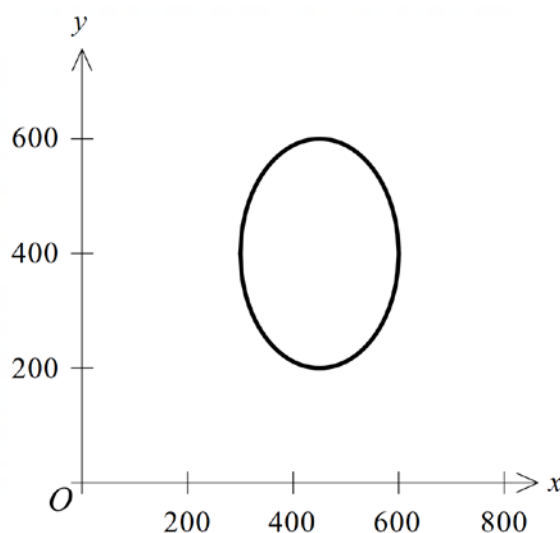
$$\frac{(x-450)^2}{22500} + \frac{(y-400)^2}{40000} = 1.$$

Question 13 continues on page 11

Question 13 (continued)

The path of the aeroplane is shown in the diagram below. At the same time that the pilot begins performing, a firework is fired from O with a velocity of 80 metres per second at an angle of inclination of θ . The position of the firework at time t relative to the fixed origin is given by $\underline{s}(t) = (80t \cos \theta) \underline{i} + (80t \sin \theta - 5t^2) \underline{j}$.

(Do NOT prove this).



- | | | |
|-------|--|----------|
| (ii) | Find the value of θ given that the firework explodes when it reaches its maximum height of 160 m. | 3 |
| | | |
| (iii) | By first finding a vector that represents the displacement of the aeroplane from the firework at time t , find how far the aeroplane is from the firework when it explodes. Give your answer to the nearest metre. | 3 |

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

(a) Use the substitution $x = \sin \theta$ to find $\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx$. **3**

(b) (i) Write $2 \sin x \sin((2k+1)x)$ as the difference of two cosine functions. **1**

(ii) Prove by mathematical induction that for all integers $n \geq 1$, **3**

$$\sin x + \sin 3x + \sin 5x + \dots + \sin(2n-1)x = \frac{1 - \cos 2nx}{2 \sin x}.$$

(c) (i) The graph of $f(x) = -\frac{1}{x^2}$ is shown on the separate Response Sheet for Question 14 c (i). **2**

On the Response Sheet, use addition of ordinates to sketch the graph of $g(x) = x^2 - \frac{1}{x^2}$ for $y \in [-10, 10]$ clearly showing the location of the x -intercepts.

You do not need to find the x -coordinates at the endpoints of the range.

(ii) Show that $g(x)$ may be rearranged to give **2**

$$x^2 = \frac{y + \sqrt{y^2 + 4}}{2}.$$

A glass with a hollow stem, and with base at $y = -10$ is made by rotating the part of $g(x)$ where $x > 0$ and $y \in [-10, 10]$ about the y -axis to form a solid of revolution, where length units are in centimetres.

(iii) Write down a definite integral which, when evaluated, would give the volume of the glass. **1**

Do not attempt to evaluate this integral.

Question 14(c) continues on page 13

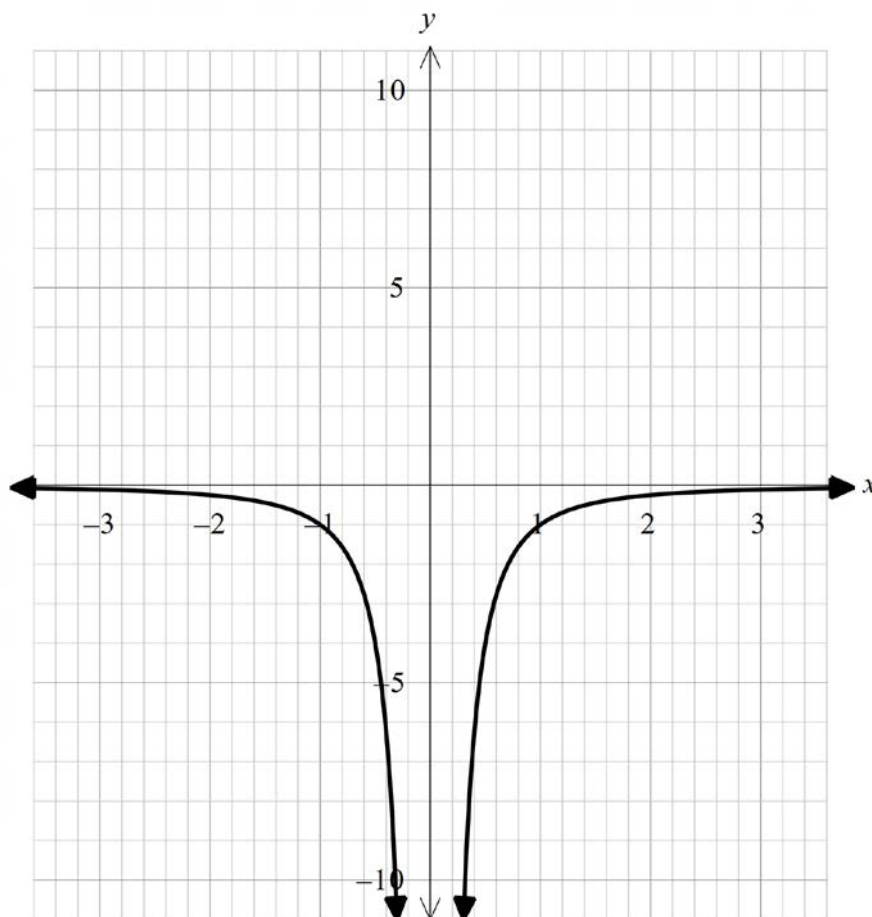
Question 14 (continued)

- (iv) Liquid is poured into the glass at a rate of 1.5 cm^3 per second. **3**
Find the rate at which the surface of the liquid is rising when it is 6 cm from the top of the glass.

End of paper

Question 14 c) (i) – Response Sheet

14 c) The graph of $f(x) = -\frac{1}{x^2}$ is shown below.



- (i) On this Response Sheet provided, use addition of ordinates to sketch the graph of the $g(x) = x^2 - \frac{1}{x^2}$ for $y \in [-10, 10]$ clearly showing the location of the x -intercepts. 2
- You do not need to find the x -coordinates at the endpoints of the range.**

Place this sheet inside your Question 14 answer booklet.

Mathematics Extension 1- Solutions

- 1 Let $P(x) = x^3 - 2ax^2 + x - 1$ where $a \in \mathbb{R}$. When $P(x)$ is divided by $x + 2$, the remainder is 5. What is the value of a ?

A. 2

B. $-\frac{7}{4}$

C. $\frac{1}{2}$

D. -2

$$\begin{aligned}
 P(x) &= x^3 - 2ax^2 + x - 1 \\
 P(-2) &= (-2)^3 - 2a(-2)^2 + (-2) - 1 \\
 &= -8 - 8a - 2 - 1 \\
 &= -8a - 11 = 5 \\
 -8a &= 16 \\
 \therefore a &= -2
 \end{aligned}$$

- 2 The points A and B have coordinates $(-2, 3)$ and $(2, -5)$ respectively.

Which of the following is the vector $\overrightarrow{AB}$?

A. $-2\hat{j}$

B. $4\hat{i} - 8\hat{j}$

C. $-4\hat{i} + 8\hat{j}$

D. $2\hat{j}$

$$\begin{aligned}
 \overrightarrow{AB} &= \underline{b} - \underline{a} \\
 &= \begin{pmatrix} 2 \\ -5 \end{pmatrix} - \begin{pmatrix} -2 \\ 3 \end{pmatrix} \\
 &= \begin{pmatrix} 4 \\ -8 \end{pmatrix}
 \end{aligned}$$

- 3 What is the angle between the vectors $\begin{pmatrix} -7 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$?

A. $\cos^{-1}(-0.8)$

B. $\cos^{-1}(-0.08)$

C. $\cos^{-1}(0.8)$

D. $\cos^{-1}(0.08)$

$$\begin{aligned}
 \cos \theta &= \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} \\
 &= \frac{\begin{pmatrix} -7 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}}{\sqrt{7^2 + 1^2} \sqrt{1^2 + 1^2}} \\
 &= \frac{-7 - 1}{\sqrt{50} \sqrt{2}} \\
 &= -\frac{8}{10} \\
 \therefore \theta &= \cos^{-1}(-0.8)
 \end{aligned}$$

4 Which of the following is the derivative of $\tan^{-1}(3x)$?

A. $3 \tan^{-1} 3x$

☒ B. $\frac{3}{1+9x^2}$

C. $\frac{3}{1+3x^2}$

D. $3 \sec^2 3x$

$$\begin{aligned} \frac{d}{dx} [\tan^{-1}(3x)] \\ &= \frac{3}{1+(3x)^2} \\ &= \frac{3}{1+9x^2} \end{aligned}$$

5 What is the equation of the inverse of $f(x) = \frac{5+e^{2x}}{3}$?

A. $y = \frac{3}{5+e^{2x}}$

B. $y = e^{5-3x}$

☒ C. $y = \frac{1}{2} \ln(3x-5)$

D. $y = \frac{1}{2} \ln(5-3x)$

$$\begin{aligned} f^{-1}: x &= \frac{5+e^{2y}}{3} \\ 3x &= 5+e^{2y} \\ e^{2y} &= 3x-5 \\ 2y &= \ln|3x-5| \\ y &= \frac{1}{2} \ln|3x-5| \end{aligned}$$

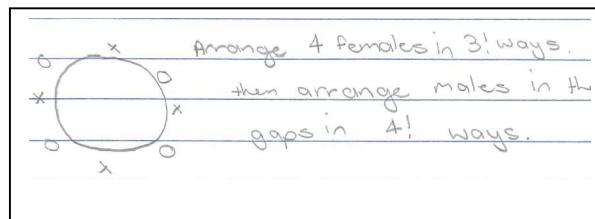
6 Four female and four male students are to be seated around a circular table. In how many ways can this be done if the males and females must alternate?

A. $4! \times 4!$

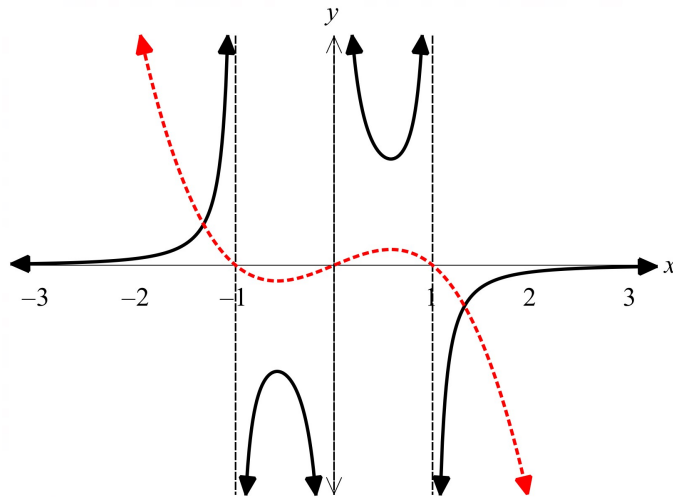
☒ B. $3! \times 4!$

C. $3! \times 3!$

D. $2 \times 3! \times 3!$



- 7 The graph below shows $y = \frac{1}{f(x)}$.



Which of the following best represents the equation of $f(x)$?

- A. $f(x) = 1 - x^2$
- B. $f(x) = x(x^2 - 1)$
- ☒ C. $f(x) = x(1 - x^2)$
- D. $f(x) = x^2(x^2 - 1)$

$$y = -x(x-1)(x+1)$$

$$= x(1-x^2)$$

- 8 What is the vector projection of $\underline{a} = 2\underline{i} + 3\underline{j}$ in the direction of $\underline{b} = \underline{i} - 4\underline{j}$?

- A. $-\frac{20}{17}\underline{i} - \frac{30}{17}\underline{j}$
- B. $-\frac{10}{13}\underline{i} + \frac{40}{13}\underline{j}$
- C. $-\frac{20}{13}\underline{i} - \frac{30}{13}\underline{j}$
- ☒ D. $-\frac{10}{17}\underline{i} + \frac{40}{17}\underline{j}$

$$\text{proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|^2} \underline{b}$$

$$= \frac{2(1) + 3(-4)}{1 + 16} \begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

$$= \frac{-10}{17} \begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

$$= -\frac{10}{17}\underline{i} + \frac{40}{17}\underline{j}$$

- 9 The radius of a sphere, r , is increasing at the rate of 0.3 cm per second.
What is the rate of increase in the volume, V , in cm^3 per second, at the instant when the surface area is $100\pi \text{ cm}^2$?

- A. 10π
B. 12π
C. 25π
D. 30π

$$\frac{dv}{dt} = \frac{dv}{dr} \times \frac{dr}{dt} \quad \text{SA} = 4\pi r^2 = 100\pi$$

$$\text{Now } V = \frac{4}{3}\pi r^3$$

$$\frac{dv}{dr} = 4\pi r^2$$

$$\frac{dv}{dt} = 100\pi \times 0.3$$

$$\frac{dv}{dt} = 30\pi$$

- 10 Which of the following is the range of the function $f(x) = |b \cos^{-1}(x) - a|$, where $a > 0$, $b > 0$ and $a < \frac{b\pi}{2}$?

- A. $[-a, b\pi - a]$
B. $[0, b\pi - a]$
C. $[a, b\pi - a]$
D. $[0, a]$

$$f(x) = |b \cos^{-1}(x) - a|$$

$$0 \leq \cos^{-1}(x) \leq \pi$$

$$0 \leq b \cos^{-1}(x) \leq b\pi$$

$$-a \leq b \cos^{-1}(x) - a \leq b\pi - a$$

$$0 \leq |b \cos^{-1}(x) - a| \leq b\pi - a$$

$$a < \frac{b\pi}{2}$$

$$2a < b\pi$$

$$0 < b\pi - 2a$$

$$\therefore b\pi - a > a > 0$$

Question 11 (15 marks)

(a) Solve $ 2x-3 \leq 1$.	2
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$$\begin{aligned} -1 &\leq 2x-3 \leq 1 \\ 2 &\leq 2x \leq 4 \\ 1 &\leq x \leq 2 \end{aligned}$$

(b) Find $\int_0^{\frac{1}{2}} \frac{dy}{\sqrt{1-3y^2}}$.	2
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$$\begin{aligned} \int_0^{\frac{1}{2}} \frac{dy}{\sqrt{1-3y^2}} &= \frac{1}{\sqrt{3}} \int_0^{\frac{1}{2}} \frac{\sqrt{3} dy}{\sqrt{1-(\sqrt{3}y)^2}} \\ &= \frac{1}{\sqrt{3}} \left[\sin^{-1}(\sqrt{3}y) \right]_0^{\frac{1}{2}} \\ &= \frac{1}{\sqrt{3}} \left(\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \sin^{-1}(0) \right) \\ &= \frac{1}{\sqrt{3}} \left(\frac{\pi}{3} - 0 \right) \\ &= \frac{\pi}{3\sqrt{3}} \end{aligned}$$

(c) Let α , β and γ be the roots of the equation $2x^3 - kx^2 - 4x + 12 = 0$.	
(i) Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.	2
(ii) Given that two of its roots sum to zero, find the third root and hence find the value of k .	2

$$\begin{aligned} \text{(i)} \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\ &= \frac{-\frac{4}{2}}{-\frac{12}{2}} \\ &= \frac{1}{3} \end{aligned}$$

ii) Let the roots be $\alpha, -\alpha, \beta$.

$$\frac{1}{\alpha} + \frac{1}{-\alpha} + \frac{1}{\beta} = \frac{1}{3} \quad \text{from part i)}$$

$$\frac{1}{\beta} = \frac{1}{3}$$

$$\beta = 3$$

$$\therefore \alpha - \alpha + 3 = \frac{k}{2} \quad (\text{sum of roots})$$

$$3 = \frac{k}{2}$$

$$k = 6$$

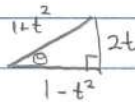
(d) Using the substitution $t = \tan \frac{\theta}{2}$, or otherwise, show that

2

$$\cot \theta + \frac{1}{2} \tan \frac{\theta}{2} = \frac{1}{2} \cot \frac{\theta}{2} \quad \text{for all } \theta \neq k\pi, k \in \mathbb{Z}.$$

$$\text{LHS} = \cot \theta + \frac{1}{2} \tan \frac{\theta}{2}$$

$$= \frac{1-t^2}{2t} + \frac{1}{2}t$$



$$= \frac{1-t^2+t^2}{2t}$$

$$= \frac{1}{2t}$$

$$= \frac{1}{2} \times \frac{1}{t}$$

$$= \frac{1}{2} \times \cot \frac{\theta}{2}$$

$$= \text{RHS}$$

(e) Find the term independent of x in the expansion of $\left(3x^2 + \frac{2}{x}\right)^{12}$.

2

$$T_k = \binom{12}{k} (3x^2)^{12-k} (2x^{-1})^k$$

$$= \binom{12}{k} 3^{12-k} x^{24-2k} \cdot 2^k x^{-k}$$

$$= \binom{12}{k} 3^{12-k} 2^k x^{24-3k}$$

For term independent of x : $24-3k=0$

$$k=8$$

$$\therefore \text{term independent of } x = \binom{12}{8} 3^4 \times 2^8$$

$$= 10204320$$

(f) Prove by mathematical induction that $n^3 + 2n$ is divisible by 3 for all positive integers n . 3

- Test for $n=1$

$$1^3 + 2(1) = 3 \quad \text{which is divisible by 3}$$

$\therefore$ The statement is true for $n=1$

- Assume the statement is true for $n=k$

$$\text{i.e. } k^3 + 2k = 3P \quad \text{for integer } P$$

- Prove for $n=k+1$

$$\text{i.e. } (k+1)^3 + 2(k+1) = 3Q \quad Q \in \mathbb{Z}$$

$$\text{LHS} = (k+1)^3 + 2(k+1)$$

$$= (k^3 + 3k^2 + 3k + 1) + 2k + 2$$

$$= (k^3 + 2k) + 3(k^2 + k + 1)$$

$$= 3P + 3(k^2 + k + 1) \quad \text{from assumption}$$

$$= 3(P + k^2 + k + 1)$$

$$= 3Q, \quad Q = P + k^2 + k + 1 \in \mathbb{Z}$$

$\therefore$ The statement is true for $n=k+1$, if it is true for $n=k$.

- By mathematical induction, it is true for all positive integers.

Question 12 (15 marks)

(a) Solve $\frac{x^2+6}{x} < 5$.

3

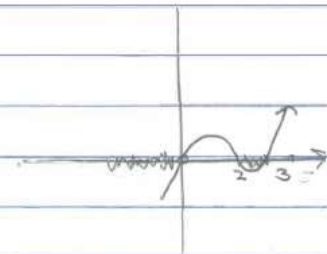
a) $\frac{x^2+6}{x} < 5$

$$(x \neq 0) \quad x(x^2+6) < 5x^2$$

$$x(x^2+6) - 5x^2 < 0$$

$$x(x^2 - 5x + 6) < 0$$

$$x(x-3)(x-2) < 0$$

$$x < 0, \quad 2 < x < 3.$$


(b) By expressing $\cos x - \sqrt{3} \sin x$ in the form $A \cos(x+\alpha)$ where $A > 0$, solve $\cos x - \sqrt{3} \sin x + 1 = 0$ for $0 \leq x \leq 2\pi$.

4

$$\cos x - \sqrt{3} \sin x = A \cos(x+\alpha)$$

$$= A \cos x \cos \alpha - A \sin x \sin \alpha$$

Equating coefficients: $A \cos \alpha = 1$ (1)

$$A \sin \alpha = \sqrt{3}$$
 (2)
$$\textcircled{1}^2 - \textcircled{2}^2: A^2 \cos^2 \alpha + A^2 \sin^2 \alpha = 1 + 3$$

$$A^2 = 4$$

$$A = 2 \quad (A > 0)$$

$\textcircled{2} \div \textcircled{1}$ $\frac{A \sin \alpha}{A \cos \alpha} = \frac{\sqrt{3}}{1}$

$$\tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

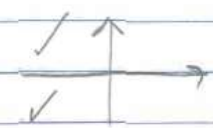
$$\therefore \cos x - \sqrt{3} \sin x = 2 \cos\left(x + \frac{\pi}{3}\right)$$

$$2 \cos\left(x + \frac{\pi}{3}\right) + 1 = 0$$

$$\cos\left(x + \frac{\pi}{3}\right) = -\frac{1}{2}$$

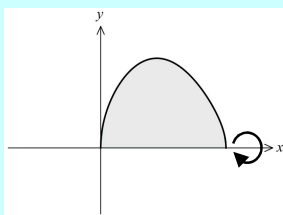
$0 \leq x \leq 2\pi$
 $\frac{\pi}{3} \leq x + \frac{\pi}{3} \leq \frac{7\pi}{3}$

$$x + \frac{\pi}{3} = \pi - \frac{\pi}{3}, \quad \pi + \frac{\pi}{3},$$

$$x = \frac{\pi}{3}, \quad \pi.$$


(c) A section of the graph of $y = \sqrt{\sin 3x \cos 2x}$ is shown in the diagram below.

4



By first finding the smallest positive solution to $\sin 3x \cos 2x = 0$, find the volume of the solid formed when the shaded region is rotated about the x -axis.

$$\sin 3x \cos 2x = 0$$

$$\sin 3x = 0$$

$$\cos 2x = 0$$

$$3x = 0, \pi, 2\pi, \dots$$

$$2x = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$x = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \dots$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \dots$$

smallest positive solution $x = \frac{\pi}{4}$

$$V = \pi \int_0^{\pi/4} \left(\sqrt{\sin 3x \cos 2x} \right)^2 dx$$

$$= \frac{\pi}{2} \int_0^{\pi/4} (\sin 5x + \sin x) dx$$

$$= \frac{\pi}{2} \left[-\frac{\cos 5x}{5} - \cos x \right]_0^{\pi/4}$$

$$= \frac{\pi}{2} \left[\left(-\frac{\cos 5\pi}{4} - \cos \frac{\pi}{4} \right) - \left(-\frac{\cos 0}{5} - \cos 0 \right) \right]$$

$$= \frac{\pi}{2} \left(\left(\frac{-1}{5\sqrt{2}} - \frac{1}{\sqrt{2}} \right) - \left(-\frac{1}{5} - 1 \right) \right)$$

$$= \frac{\pi}{2} \left(\frac{1 - 5 + \sqrt{2} + 5\sqrt{2}}{5\sqrt{2}} \right)$$

$$= \frac{\pi}{10\sqrt{2}} (6\sqrt{2} - 4)$$

$$= \frac{\pi}{5\sqrt{2}} (3\sqrt{2} - 2)$$

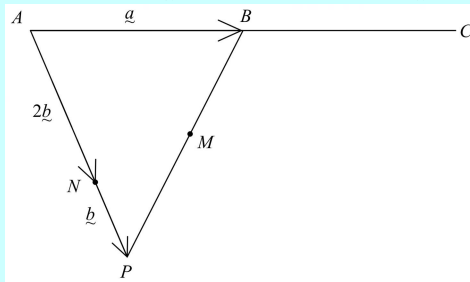
$$= \frac{\pi}{10} (6 - 2\sqrt{2}) \text{ u}^3$$

(d) In the diagram below APB is a triangle. N is a point on AP .

$$\overrightarrow{AB} = \underline{a}$$

$$\overrightarrow{AN} = 2\underline{b}$$

$$\overrightarrow{NP} = \underline{b}$$



(i) Find the vector $\overrightarrow{PB}$ in terms of $\underline{a}$ and $\underline{b}$.

1

(ii) B is the midpoint of AC . M is the midpoint of PB .

3

Show that NMC is a straight line.

$$\begin{aligned} \text{(i)} \quad \overrightarrow{PB} &= \overrightarrow{PA} + \overrightarrow{AB} \\ &= -3\underline{b} + \underline{a} \\ &= \underline{a} - 3\underline{b} \end{aligned}$$

$$\text{(ii)} \quad \overrightarrow{AC} = 2\underline{a}$$

$$\begin{aligned} \overrightarrow{PM} &= \frac{1}{2} \overrightarrow{PB} \\ &= \frac{1}{2} (\underline{a} - 3\underline{b}) \end{aligned}$$

$$\begin{aligned} \overrightarrow{NM} &= \overrightarrow{PM} - \overrightarrow{PN} \\ &= \frac{1}{2} (\underline{a} - 3\underline{b}) - (-\underline{b}) \\ &= \frac{1}{2} (\underline{a} - \underline{b}) \end{aligned}$$

$$\begin{aligned} \overrightarrow{NC} &= \overrightarrow{NA} + \overrightarrow{AC} \\ &= -2\underline{b} + 2\underline{a} \\ &= 2(\underline{a} - \underline{b}) \end{aligned}$$

$$\therefore \overrightarrow{NC} = 4 \overrightarrow{NM}$$

$\overrightarrow{NC}$ is a scalar multiple of $\overrightarrow{NM}$ and so NMC is a straight line.

Question 13 (15 marks)

- (a) A netball team's record for the 2022 season was 16 wins and 4 losses. 2
 None of their games were drawn. Prove that the team must have won at least 4 games in a row somewhere during the season.

There are 5 spots to place the wins between the 4 losses in the season

— L — L — L — L —

$\therefore$ There are 5 categories to place the 16 objects (wins).

$$\frac{16}{5} = 3.2$$

$\therefore$ There must be at least 4 wins in one of the categories.

By the pigeonhole principle the team must have won 4 games in a row somewhere during the season.

- (b) The letters of the word REORDER are arranged randomly in a line.

- (i) Use a combinatorial argument to explain why 2

$$\binom{7}{3} \binom{4}{2} \binom{2}{1} \binom{1}{1} = \binom{7}{1} \binom{6}{1} \binom{5}{2} \binom{3}{3}.$$

- (ii) Hence, or otherwise, find the probability that a random rearrangement has all the consonants grouped together. 3

R R R

E E 7 letters

O

D

To make an arrangement of these letters:

choose 3 spots for the R's in $\binom{7}{3}$ ways

then choose 2 spots for the E's from the remaining 4 spots

that in $\binom{4}{2}$ ways.

then choose the spot for the O from the remaining 2 spots

in $\binom{2}{1}$ ways

and then finally place the D in $\binom{1}{1}$ ways.

$$\therefore \text{total \# of arrangements} = \binom{7}{3} \binom{4}{2} \binom{2}{1} \binom{1}{1}$$

Alternately first place the D in $\binom{7}{1}$ ways,
 then place the O in $\binom{6}{1}$ ways,
 then place the Es in $\binom{5}{2}$ ways
 and finally place the Rs in $\binom{3}{3}$ ways.

$\therefore$ total # of arrangements $\binom{7}{1}\binom{6}{1}\binom{5}{2}\binom{3}{3}$

$$\therefore \binom{7}{3}\binom{4}{2}\binom{2}{1}\binom{1}{1} = \binom{7}{1}\binom{6}{1}\binom{5}{2}\binom{3}{3}$$

ii) Arrangements with consonants grouped together:

RRRD OEE



4 groups

Arrange groups $\frac{4!}{2!}$ ways

and then arrange consonants in $\frac{4!}{3!}$ ways

$$P(\text{consonants grouped together}) = \frac{4!}{2!} \times \frac{4!}{3!}$$

$$\frac{\binom{7}{3}\binom{4}{2}\binom{2}{1}\binom{1}{1}}{35} \leftarrow \text{from (i)}$$

$$= \frac{4}{35}$$

- (c) A pilot is performing an air show. The position of her aeroplane at time t relative to a fixed origin O is given by $\underline{r}(t) = \left(450 - 150 \sin\left(\frac{\pi t}{6}\right)\right)\underline{i} + \left(400 - 200 \cos\left(\frac{\pi t}{6}\right)\right)\underline{j}$, where $\underline{i}$ is a unit vector in a horizontal direction and $\underline{j}$ is a unit vector vertically up. Displacement components are measured in metres and time t is measured in seconds where $t \geq 0$.

- (i) Show that the cartesian equation of the path of the aeroplane is given by:

2

$$\frac{(x-450)^2}{22500} + \frac{(y-400)^2}{40000} = 1.$$

c) (i) $\underline{r}(t) = \begin{bmatrix} 450 - 150 \sin\left(\frac{\pi t}{6}\right) \\ 400 - 200 \cos\left(\frac{\pi t}{6}\right) \end{bmatrix}$

$x = 450 - 150 \sin\left(\frac{\pi t}{6}\right) \Rightarrow \frac{x-450}{-150} = \sin\left(\frac{\pi t}{6}\right) \quad (1)$

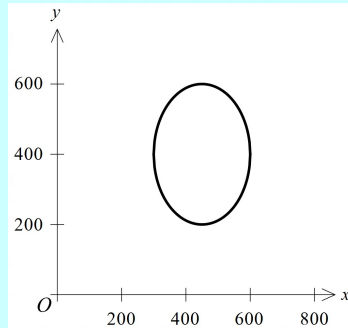
$y = 400 - 200 \cos\left(\frac{\pi t}{6}\right) \Rightarrow \frac{y-400}{-200} = \cos\left(\frac{\pi t}{6}\right) \quad (2)$

$(1)^2 + (2)^2 \quad \frac{(x-450)^2}{22500} + \frac{(y-400)^2}{40000} = \sin^2\left(\frac{\pi t}{6}\right) + \cos^2\left(\frac{\pi t}{6}\right)$

$\frac{(x-450)^2}{22500} + \frac{(y-400)^2}{40000} = 1.$

The path of the aeroplane is shown in the diagram below. At the same time that the pilot begins performing, a firework is fired from O with a velocity of 80 metres per second at an angle of inclination of θ . The position of the firework at time t relative to the fixed origin is given by $\underline{s}(t) = (80t \cos \theta)\underline{i} + (80t \sin \theta - 5t^2)\underline{j}$.

(Do NOT prove this).



- (ii) Find the value of θ given that the firework explodes when it reaches its maximum height of 160 m. 3
- (iii) By first finding a vector that represents the displacement of the aeroplane from the firework at time t , find how far the aeroplane is from the firework when it explodes. Give your answer to the nearest metre. 3

$$\text{ii) } \underline{s}(t) = \begin{bmatrix} 80t \cos \theta \\ 80t \sin \theta - 5t^2 \end{bmatrix}$$

$$s_y = 80t \sin \theta - 5t^2$$

$$\dot{y} = 80 \sin \theta - 10t$$

$$\text{when } \dot{y} = 0, \quad 80 \sin \theta - 10t = 0$$

$$t = 8 \sin \theta$$

$$\text{when } t = 8 \sin \theta, \quad y = 160$$

$$80(8 \sin \theta) \sin \theta - 5(8 \sin \theta)^2 = 160$$

$$640 \sin^2 \theta - 320 \sin^2 \theta = 160$$

$$320 \sin^2 \theta = 160$$

$$\sin^2 \theta = \frac{1}{2}$$

$$\sin \theta = \pm \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4} \quad \left(0 \leq \theta \leq \frac{\pi}{2}\right)$$

$$\text{iii) } \vec{SR} = \vec{OR} - \vec{OS}$$

$$= \begin{bmatrix} 450 - 150 \sin\left(\frac{\pi t}{6}\right) \\ 400 - 200 \cos\left(\frac{\pi t}{6}\right) \end{bmatrix} - \begin{bmatrix} 80t \cos \theta \\ 80t \sin \theta - 5t^2 \end{bmatrix}$$

$$\theta = \frac{4\pi}{4} \Rightarrow \sin \theta = \cos \theta = \frac{1}{\sqrt{2}}$$

$$t = 8 \sin \theta \\ = 8/\sqrt{2}$$

$$\vec{SR} = \begin{bmatrix} 450 - 150 \sin\left(\frac{\pi \times \frac{8}{\sqrt{2}}}{6}\right) - 80\left(\frac{8}{\sqrt{2}}\right) \times \frac{1}{\sqrt{2}} \\ 400 - 200 \cos\left(\frac{\pi \times \frac{8}{\sqrt{2}}}{6}\right) - 80\left(\frac{8}{\sqrt{2}}\right) \times \frac{1}{\sqrt{2}} + 5\left(\frac{8}{\sqrt{2}}\right)^2 \end{bmatrix}$$

$$= \begin{bmatrix} 450 - 150 \sin\left(\frac{2\sqrt{2}\pi}{3}\right) - 320 \\ 400 - 200 \cos\left(\frac{2\sqrt{2}\pi}{3}\right) - 320 + 160 \end{bmatrix}$$

$$= \begin{bmatrix} 130 - 150 \sin\left(\frac{2\sqrt{2}\pi}{3}\right) \\ 240 - 200 \cos\left(\frac{2\sqrt{2}\pi}{3}\right) \end{bmatrix}$$

$$|\vec{SR}| = \sqrt{\left(130 - 150 \sin\left(\frac{2\sqrt{2}\pi}{3}\right)\right)^2 + \left(240 - 200 \cos\left(\frac{2\sqrt{2}\pi}{3}\right)\right)^2}$$

$$= \sqrt{201426.25 \dots}$$

$$= 448.8 \text{ m}$$

$$= 449 \text{ m (nearest metre)}$$

Question 14 (15 marks)

(a) Use the substitution $x = \sin \theta$ to find $\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx$.

3

$$\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx.$$

$$\text{Let } x = \sin \theta \Rightarrow \theta = \sin^{-1}(x)$$

$$dx = \cos \theta d\theta.$$

$$\text{when } x = 0, \theta = 0$$

$$\text{when } x = \frac{1}{2}, \theta = \frac{\pi}{6}.$$

$$= \int_0^{\pi/6} \frac{\sin^2 \theta \cos \theta d\theta}{\sqrt{1-\sin^2 \theta}}$$

$$= \int_0^{\pi/6} \frac{\sin^2 \theta \cos \theta d\theta}{\sqrt{\cos^2 \theta}}$$

$$= \int_0^{\pi/6} \frac{\sin^2 \theta \cancel{\cos \theta} d\theta}{\cancel{\cos \theta}} \quad \cos \theta > 0$$

$$= \int_0^{\pi/6} \frac{1}{2} (1 - \cos 2\theta) d\theta.$$

$$= \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^{\pi/6}$$

$$= \left[\left(\frac{\pi}{12} - \frac{1}{4} \sin \left(\frac{\pi}{3} \right) \right) - (0 - 0) \right]$$

$$= \frac{\pi}{12} - \frac{\sqrt{3}}{8} = \frac{2\pi - 3\sqrt{3}}{24}$$

(b)	(i)	Write $2 \sin x \sin((2k+1)x)$ as the difference of two cosine functions.	1
	(ii)	Prove by mathematical induction that for all integers $n \geq 1$, $\sin x + \sin 3x + \sin 5x + \dots + \sin(2n-1)x = \frac{1 - \cos 2nx}{2 \sin x}.$	3

$$\begin{aligned}
 \text{i)} \quad 2 \sin x \sin((2k+1)x) &= \cos(x - (2k+1)x) - \cos(x + (2k+1)x) \\
 &= \cos(-2kx) - \cos(2(k+1)x) \\
 &= \cos(2kx) - \cos(2(k+1)x) \quad (\cos \text{ is even fn})
 \end{aligned}$$

$$\sin x + \sin 3x + \sin 5x + \dots + \sin (2n-1)x = \frac{1 - \cos 2nx}{2 \sin x}$$

• Test for $n=1$

$$\begin{aligned} \text{LHS} &= \sin (2(1)-1)x \\ &= \sin x \end{aligned}$$

$$\text{RHS} = \frac{1 - \cos 2(1)x}{2 \sin x}$$

$$= \frac{1 - \cos 2x}{2 \sin x}$$

$$= \frac{2 \sin^2 x}{2 \sin x}$$

$$= \sin x$$

$\therefore$ True for $n=1$

• Assume true for $n=k$

$$\text{i.e. } \sin x + \sin 3x + \sin 5x + \dots + \sin (2k-1)x = \frac{1 - \cos 2kx}{2 \sin x}$$

• Prove true for $n=k+1$

$$\text{i.e. } \sin x + \sin 3x + \sin 5x + \dots + \sin (2(k+1)-1)x = \frac{1 - \cos 2(k+1)x}{2 \sin x}$$

$$\text{LHS} = \sin x + \sin 3x + \sin 5x + \dots + \sin (2k-1)x + \sin (2(k+1)-1)x$$

$$= \frac{1 - \cos 2kx}{2 \sin x} + \sin (2(k+1)-1)x \quad \text{using assumption}$$

$$= \frac{1 - \cos 2kx}{2 \sin x} + \frac{2 \sin x \sin (2k+1)x}{2 \sin x}$$

$$= \frac{1 - \cancel{\cos 2kx} + \cancel{\cos (2kx)} - \cos 2(k+1)x}{2 \sin x} \quad \text{from part i,}$$

$$= \frac{1 - \cos 2(k+1)x}{2 \sin x} = \text{RHS}$$

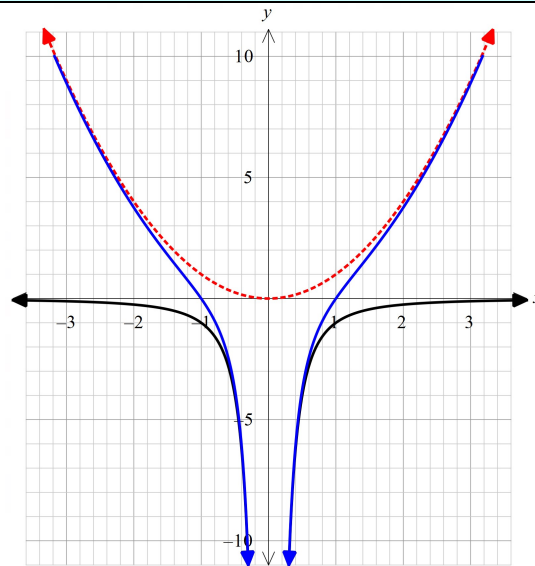
$\therefore$ By mathematical induction it is true for all integers $n \geq 1$

(c) (i) The graph of $f(x) = -\frac{1}{x^2}$ is shown below.

2

Use addition of ordinates to sketch the graph of $g(x) = x^2 - \frac{1}{x^2}$ for $y \in [-10, 10]$ clearly showing the location of the x -intercepts.

You do not need to find the x -coordinates at the endpoints of the range.



(ii) Show that $g(x)$ may be rearranged to give $x^2 = \frac{y + \sqrt{y^2 + 4}}{2}$.

2

$$g(x) = x^2 - \frac{1}{x^2}$$

$$y = x^2 - \frac{1}{x^2}$$

$$yx^2 = x^4 - 1 \Rightarrow x^4 - x^2y - 1 = 0$$

$$x^2 = \frac{y \pm \sqrt{(-y)^2 - 4(1)(-1)}}{2}$$

$$= \frac{y \pm \sqrt{y^2 + 4}}{2}$$

$$\therefore x^2 = \frac{y + \sqrt{y^2 + 4}}{2} \quad (x^2 > 0)$$

A glass with a hollow stem, and with base at $y = -10$ is made by rotating the part of $g(x)$ where $x > 0$ and $y \in [-10, 10]$ about the y -axis to form a solid of revolution, where length units are in centimetres.

- (iii) Write down a definite integral which, when evaluated, would give the volume of the glass. 1
- (iv) Liquid is poured into the glass at a rate of 1.5 cm^3 per second. 3
Find the rate at which the surface of the liquid is rising when it is 6 cm from the top of the glass.

$$\text{iii)} \quad V = \pi \int_{-10}^{10} \frac{y + \sqrt{y^2 + 4}}{2} dy$$

$$\text{iv)} \quad \frac{dv}{dt} = 1.5. \quad \text{Find } \frac{dy}{dt} \text{ when } y = 4 \quad (\text{top of glass is } y=10)$$

$$V(y) = \int_{-10}^y \pi \left(\frac{y + \sqrt{y^2 + 4}}{2} \right) dy = \int_{-10}^y \frac{dv}{dy} dy$$

$$\frac{dv}{dt} = \frac{dv}{dy} \times \frac{dy}{dt}$$

$$1.5 = \pi \left(\frac{y + \sqrt{y^2 + 4}}{2} \right) \times \frac{dy}{dt}$$

$$\text{At } y = 4$$

$$1.5 = \pi \left(\frac{4 + \sqrt{16 + 4}}{2} \right) \times \frac{dy}{dt}$$

$$\frac{dy}{dt} = \frac{1.5 \times 2}{\pi (4 + \sqrt{20})} = 0.1127 \dots$$

Rising at a rate of 0.1 cm/s